

Class 23 Fields

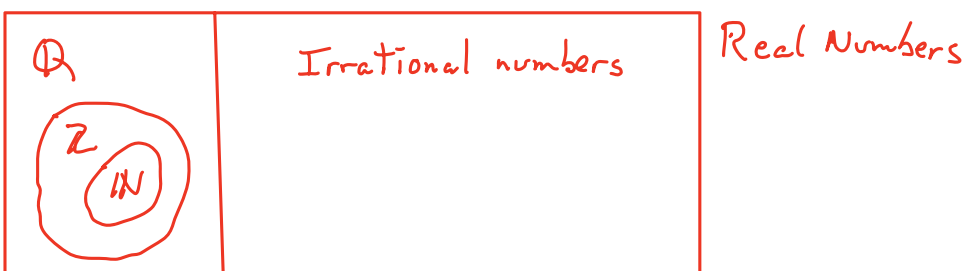
Goals

1. Understand the definition of a field.
2. Be able to prove simple properties of fields.
3. Develop familiarity with examples.

Definition: A field is a nonempty set, F , with two binary operations called addition and multiplication satisfying 3 properties:

1. F is an abelian group under addition.
2. F^* (nonzero elements) is an abelian group under multiplication.
3. Multiplication distributes over addition
 $a \cdot (b+c) = (a \cdot b) + (a \cdot c)$ for all a, b, c in F .

Examples The set of Rational numbers, $\mathbb{Q}$, form a field.
The set of real numbers, $\mathbb{R}$, form a field.
The set $\mathbb{Z}_p$ is a field iff p is prime
The set of integers, $\mathbb{Z}$, is not



Real numbers - rational and irrational

Defining real numbers rigorously is difficult.

Was not done until 1800's.

Any real number is the limit of a sequence of rational numbers.

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \quad \frac{1}{2} = \lim_{n \rightarrow \infty} \frac{1}{2}$$

We commonly think of decimal representation for a real number.

$$\frac{3}{4} = .75 \quad \frac{5}{3} = 1.66\overline{6} \quad \pi \approx 3.14$$

Rational numbers have decimal representations that repeat periodically. Irrational numbers have decimal representations that are nonrepeating.

Practice: Write each repeating decimal as a ratio of integers.

a. $0.\overline{123} = x$ This means $1000x = 123.\overline{123}$
 $1000x - x = 123.$ so $999x = 123.$ so $x = \frac{123}{999}$

b. $7.29595\overline{95} = x$ This means $10x = 72.\overline{95}$
and $1000x = 7295.\overline{95}$.
so $1000x - 10x = 7223.$ so $x = \frac{7223}{990}$

Properties of fields

How do the two operations in a field interact? For example, 0 is the additive identity. How does it work with multiplication?

Use the distributive property and cancellation to prove $0 \cdot x = 0 \quad \forall x \text{ in } F$.

Let x be any element of a field F .

$$0 + \cancel{0 \cdot x} = 0 \cdot x = (0 + 0) \cdot x = 0 \cdot x + \cancel{0 \cdot x}.$$

Using right cancellation we get

$$0 = 0 \cdot x$$

Similar Problem: If -1 is the additive inverse of the multiplicative identity in F , and x is any element of F , show that the product of -1 with x produces the additive inverse of x .

Homework:

Prove: If x is any element of a field F , then $-1 \cdot x = -x$.

Prove that $\mathbf{Z}_6$ is not a field.